Capstone Project Report: Autonomous AI Research Agent

1. Project Overview

This project provides a robust solution to the literature overload faced by modern researchers. By automating the pipeline of finding, reading, and extracting insights from highly technical documents, this autonomous agent saves significant time. The system accepts unstructured domain keywords, queries open archives, downloads full-text documentation, and synthesizes structured architectural intelligence using an LLM.

2. Architecture & Workflow Design

The technical core uses a decoupled 4-stage processing pipeline:

• **Ingestion Interface:** Dynamic parameters captured via a modular interface (Streamlit UI).

• **Discovery Agent:** Programmatic search executed via the arXiv API wrapper.

• **Document Processing:** Server-side multi-page PDF parsing via PyPDFLoader to generate clean token sequences.

• **Synthesis Engine:** Map-Reduce context windowing chain powered by LangChain and OpenAI models to assemble coherent structural summaries.

3. Core Technical Metrics Matrix

Framework Element	Selected Technology	Primary Architectural Responsibility
Orchestration	LangChain Framework	State management, Prompt Templates, Map-Reduce Execution
Discovery Layer	arXiv Search Interface	Remote automated semantic paper ingestion
Synthesis LLM	GPT-3.5-Turbo (16k)	Token mapping, structural summarization, and cognitive reasoning
Presentation	Streamlit Dashboard	End-user parameters capture and reactive rendered summaries

4. Implementation Challenges & Mitigations

• **Context Window Restrictions:** Academic papers often exceed default model sequence capacities. *Mitigation:* Implemented a split-map-reduce processing topology inside LangChain.

• **API Rate Penalties:** Rapid sequential lookup actions risk structural blocks. *Mitigation:* Configured automatic request throttling windows inside the search iteration sequence.

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